

STEPHEN NYARANGA

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A diligent young professional looking to apply the knowledge acquired at school to deliver value to the organization and for personal career growth and development.

EDUCATION

PhD in Climate Change Science

AIMS Research & Innovation Centre (AIMS-RIC) / JKUAT (Dept. of Geomatic Engineering & GIS) / Imperial College London

Oct 2025 -- Present

- CLARITY-Africa Doctoral Programme
- Integrating Deep Learning and SAR Analysis for Flood Extent Mapping, Depth Estimation, and Urban Resilience in Musanze and Kigali, Rwanda.
- Host University: Jomo Kenyatta University of Agriculture and Technology (JKUAT), Department of Geomatic Engineering and Geospatial Information Systems.

MSc Geophysics (Pursued)

ICTP-East Africa Institute for Fundamental Research (ICTP-EAIFR), University of Rwanda, Kigali

Jan 2023 -- Dec 2024

- Thesis: Time Series InSAR Analysis over the Olkaria Geothermal Power Plant in Kenya: Insights From a 2020--2024 Sentinel-1 InSAR Survey.
- Coursework: Seismology, Geodesy, InSAR, Computational Geodynamics, Earthquake Mechanics, Atmospheric Physics, GPS Geodesy, Physics of Volcanoes, Theoretical & Observational Seismology.

MSc Mathematical Sciences -- Climate Science / Data Science

African Institute for Mathematical Sciences (AIMS-Senegal), Mbour, Senegal

Oct 2021 -- Jul 2022

- Graduated with Distinction. Mastercard Foundation Scholar.
- Dissertation: Atmospheric Rivers and Surface Melting over the Pine Island Glacier in 1983 and 1998.
- Coursework: Machine Learning, AI, Climate Dynamics, Data Science, Quantum Mechanics & Quantum Computing, Statistical Learning, Classical Mechanics.

BSc Education Science (Physics / Mathematics)

Karatina University, Nyeri, Kenya

Aug 2016 -- Mar 2021

- Graduated with First Class Honours. Registered with Teachers Service Commission (TSC), Kenya.
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PROFESSIONAL EXPERIENCE

Assistant Lecturer (Part-Time)

Vatel Rwanda, Kigali, Rwanda

Jun 2025 -- Oct 2025

- Appointed to academic rank of Assistant Lecturer for the 2025--2026 academic year.
- Delivered ICDL courses to lecturers, staff, and students.
- Led IT Department as Head of Department.
- Facilitated AI integration in curriculum and staff development.

Head of IT Department (Full-Time)

Vatel Rwanda, Kigali, Rwanda

Jul 2024 -- Oct 2025

- Contributed to academic quality assurance and digital transformation of teaching.
- Designed and developed the Learning Management System and Data Controller.
- Managed Canvas, Moodle, and Triade LMS customization.
- Built in-house platforms for tracking internships, lecturers, hours, and attendance.
- Provided IT support, training, troubleshooting, and orientation for all staff and students.

IT Associate Intern -- Mastercard Foundation Graduate Enrichment Program

Vatel Rwanda, Kigali, Rwanda

Jul 2023 -- Jul 2024

- Introduced AI tools into teaching methodologies; transitioned to Canvas Instructure.

- Designed training programs on AI applications, ICDL modules, and digital literacy.
- Managed IT systems ensuring uninterrupted services and resolving technical challenges.

Physics & Mathematics Teacher

Kathiani Boy's High School, Machakos / Ntangi Secondary School, Meru, Kenya

Jan 2021 -- Oct 2021

- Prepared lessons, teaching materials, and assessed students' progress in Mathematics and Physics.

GRADUATE SUPERVISION

Co-supervisor -- AIMS Rwanda (2026)

- Student: Richard Usengimana
- Spatial Analysis of Child Malnutrition in Rwanda: District-Level Integration of DHS Data and Agro-Climatic Indicators.
- Primary supervisor: Dr. Lema Seknewna (AIMS-RIC).

SELECTED PROJECTS

TembeaAfrica AI -- AI-Powered Travel Assistant for Africa

AIMS-RIC Doctoral Training School

2026

- Led Team 5 to build an AI-powered multi-lingual travel assistant for Africa.
- Won Best Project Award. Deployed at tembeafrica.github.io.

HeatAlert Accra -- SolarCool CoolTree Early Warning System

Imperial-AIMS Global Fellows Programme, Accra, Ghana

Jun 2026

- Team Lead (Team PUM-CLID). LSTM-based WBGT heat forecast engine with CoolShade (solar PV-powered PCM cooling shelter) and CoolBand (elastocaloric Nitinol wearable).
- Deployed HeatAlert dashboard on Hugging Face Spaces.
- Recognised as Best Innovative Accelerator Project.

Seismic Early Warning Application

Geodesy for a Resilient Africa Summer School, AIMS-Cameroon, Limbe

Jun--Jul 2026

- Seismic data visualization and alerting web app. Deployed at seis-ews-five.vercel.app.

AXIOM Research Group Website

AIMS-RIC, Kigali

Current

- Designed and built the website for the AXIOM Research Group (axiomresearchs.org).

Object Detection Demo for High Schools

STEM Outreach Project

2025

- Real-time item detection system developed as an educational STEM outreach demo for secondary school students.

CONFERENCES, WORKSHOPS & TRAINING

Geodesy for a Resilient Africa Summer School

AIMS-Cameroon, Limbe, Cameroon

Jun 29 -- Jul 5, 2026

Imperial-AIMS Global Fellows Programme -- Accra

Team Lead (PUM-CLID); Best Innovative Accelerator

Jun 1--6, 2026

EarthScope: InSAR Processing and Theory (GMTSAR)

Selected from 300+ applicants

Aug 3--7, 2026

EarthScope Seismology Skill Building Workshop (SSBW)

70-hour virtual course; earned Linux & GMT micro-credential

Jun 8 -- Aug 28, 2026

AIMS-RIC Training School: Data Science Methods

Oral presentation on SAR-Based Flood Mapping in Rwanda

Mar--Apr 2026

NASA ARSET: Fundamentals of Remote Sensing

Certificate awarded Oct 30, 2025

Oct 2025

Ministry of Education, Rwanda -- Data Preparation & Analysis

Data controllers for tertiary-level education

Sep 30 -- Oct 4, 2024

ICDL Certification & Trainer

Jul -- Nov 2023

Cybersecurity, AI, MS Office, Big Data, ICT in Education

ICTP-EAIFR-IUGG Workshop on Computational Geodynamics

Building New Expertise Across Africa

Jul 2--7, 2023

7th ASESMA (Electronic Structure Methods)

Jun 12--23, 2023

Jobberman's Finishing School Program

Mastercard Foundation partnership

May 8--19, 2023

African School of Fundamental Physics (ASP)

Renewable energies, HEP, Astronomy, Cosmology

Nov--Dec 2022

AWARDS, SCHOLARSHIPS & HONOURS

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| • Imperial-AIMS Global Fellows -- Best Innovative Accelerator Project (Team PUM-CLID) | <i>Jun 2026</i> |
| • AIMS-RIC Doctoral Training School -- Best Project Award (TembeaAfrica AI) | <i>2026</i> |
| • EarthScope / Miami University -- Intro Linux & GMT Micro-Credential Badge | <i>Jun 2026</i> |
| • ICTP-EAIFR -- Best 3-Minute Thesis Award (PhD & Master's Competition) | <i>Jul 2024</i> |
| • Mastercard Foundation Scholarship -- MSc at AIMS-Senegal | <i>Oct 2021 -- Jul 2022</i> |
| • NASA ARSET -- Fundamentals of Remote Sensing Certificate | <i>Oct 2025</i> |
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SKILLS & INTERESTS

Programming: Python, R, C/C++, MATLAB, Fortran, Linux Command, LaTeX, Microsoft Office

Data Science & ML: Deep Learning, Machine Learning, AI, LSTM, Computer Vision, Statistical Learning

Earth Observation: SAR/InSAR, Optical EO, GPS Geodesy, GMTSAR, Remote Sensing, Climate Dynamics

IT & EdTech: Canvas LMS, Moodle, Web Development, Digital Transformation, ICDL Training

Tools: Git, SharePoint, OneDrive, Hugging Face Spaces

Languages: English (Fluent), Swahili (Fluent), Luhya (Fluent)

Soft Skills: Excellent Communicator, Complex Problem Solver, Innovator, Service Oriented

MEMBERSHIPS

- AXIOM Research Group -- AIMS-RIC
- Teachers Service Commission (TSC) -- Kenya
- African School of Fundamental Physics and Applications (ASP)
- African Strategy for Fundamental and Applied Physics (ASFAP)